

Chapter-04

1. Attributes in Computer Graphics

Definition: An attribute is a property that determines how a graphical object appears on the display screen. Attributes do not change the object's shape or position; instead, they control its visual appearance.

Examples

- Color
- Line thickness
- Line style
- Character font
- Character size
- Curve style
- Fill pattern

Types of Attributes

- Line Attributes
- Curve Attributes
- Area-Fill Attributes
- Character Attributes
- Color Attributes

Why are Attributes Used?

Attributes help to

- Improve the appearance of graphics.
- Distinguish different objects.
- Highlight important information.
- Produce realistic graphical output.
- Make drawings easier to understand.

How Attributes are Specified

According to the textbook, graphics systems generally use **two methods**:

1. Attribute Parameters

Every graphics function contains additional parameters.

Example: `DrawLine(x1, y1, x2, y2, Color, Width)`

2. Current Attribute List

The graphics system stores current attribute values separately.

Example

`SetLineColor(Blue)`

`SetLineWidth(3)`

`DrawLine(...)`

`DrawLine(...)`

All subsequent lines are drawn using the current attribute values until they are changed. This approach is used in standards such as **GKS** and **PHIGS**.

2. Line Attributes

Definition: Line attributes determine how a line is displayed on the screen. A line can have different:

- Style
- Width
- Color

A. Line Type (Line Style)

The appearance of a line depends on its style.

Common Line Styles

Line Type	Description
Solid Line	Continuous line without breaks
Dashed Line	Line made of long dashes
Dotted Line	Line made of dots
Dash-Dot Line	Combination of dashes and dots

How Dashed Lines are Generated

Raster graphics systems generate dashed lines using a **pixel mask**.

Example mask

1111000

Meaning

1111 → Draw pixels

000 → Skip pixels

This produces



A pixel mask is simply a sequence of **1s** (draw) and **0s** (skip).

Problem with Fixed Pixel Masks

If the same number of pixels is used,

Horizontal dash



Diagonal dash



The diagonal dash becomes longer because of the line slope.

To keep dash lengths approximately equal:

- Adjust pixel count according to slope, or
- Draw each dash as a separate short line segment.

B. Line Width

Determines the thickness of the line. The width is controlled by a positive parameter.

1 = Standard width

2 = Double width

3 = Triple width

Values greater than 1 produce thicker lines.

Examples

Thin Line



Medium Line



Thick Line



Thick Line Generation

For raster graphics:

If slope < 1

Draw **vertical pixel spans**



If slope > 1

Draw **horizontal pixel spans**



This creates a thicker appearance.

C. Line Caps

When thick lines end, their endpoints are called **caps**.

D. Line Color

Specifies the color used to draw the line.

Examples

- Black
- Red
- Green
- Blue
- Yellow
- Cyan
- Magenta
- White

Applications

- Engineering drawing
- Road maps
- CAD software
- Graph plotting

3. Curve Attributes

Definition: Curve attributes determine the visual appearance of curves. They are almost identical to line attributes.

Curve attributes include:

- Curve color
- Curve width
- Curve style

Curve Styles

Style	Description
Solid Curve	Continuous curve
Dashed Curve	Broken curve
Dotted Curve	Dot pattern
Thick Curve	Larger width

Example

Solid

~~~~~

Dashed

~ ~ ~ ~ ~

Dotted

. . . . .

### 4. Color and Grayscale Levels

#### Definition

Color and Grayscale Levels determine how different colors or shades of gray are represented and displayed on a graphics system. The available colors depend on the display hardware and the number of bits used per pixel in the frame buffer.

#### 4. 1. Color Levels

##### What is a Color Level?

A color level is a specific color that can be displayed on the screen. Each color is represented by a numerical code.

- Color codes are integers (0, 1, 2, ...).
- In CRT monitors, these codes are converted into intensity values for the Red (R), Green (G), and Blue (B) electron guns.

#### 4.2. Color Representation in a Frame Buffer

A frame buffer stores the color information of every pixel. There are two methods of storing color information:

##### (A) Direct Color Storage

- Color codes are stored directly in the frame buffer.
- Each pixel contains its own RGB value.
- More bits per pixel → More colors available.

### Example

| Bits per Pixel | Number of Colors                 |
|----------------|----------------------------------|
| 3 bits         | 8 colors                         |
| 6 bits         | 64 colors                        |
| 24 bits        | 16.7 million colors (True Color) |

### RGB Color Model

RGB stands for:

- **R** = Red
- **G** = Green
- **B** = Blue

Each bit controls the intensity of one color gun.

#### 3-Bit RGB Color Table

| Binary | Color   |
|--------|---------|
| 000    | Black   |
| 001    | Blue    |
| 010    | Green   |
| 011    | Cyan    |
| 100    | Red     |
| 101    | Magenta |
| 110    | Yellow  |
| 111    | White   |

This is the minimum RGB color representation using 3 bits per pixel.

### B. Color Lookup Table (Color Table)

Instead of storing the full RGB value for every pixel, the frame buffer stores only an index. That index points to a Color Lookup Table (CLUT) containing the actual RGB values.

#### Working

Frame Buffer

|



Color Index

|



Color Lookup Table

|



RGB Value

|



Display

Example:

- Pixel value = **196**
- Lookup table entry 196 stores RGB value **2081**
- That RGB value is displayed on the screen.

### Advantages of Color Tables

- Requires less frame-buffer memory
- Supports many colors with a small buffer
- Colors can be changed by modifying only the lookup table
- Useful for visualization and image processing
- Allows threshold coloring without changing pixel values

## 5. Grayscale Levels

### Definition

A grayscale image contains only shades of gray instead of colors. Gray scale consists of shades ranging from black to white. It is mainly used on monochrome displays.

### Gray Color

A gray color is obtained when  $R = G = B$ . That means all three RGB components have the same intensity.

Example

(100,100,100)

(200,200,200)

(50,50,50)

### Intensity Range

Intensity values vary between

0 → Black

1 → White

Values between 0 and 1 produce different gray shades.

### Four-Level Grayscale System

| Intensity | Binary | Display    |
|-----------|--------|------------|
| 0.00      | 00     | Black      |
| 0.33      | 01     | Dark Gray  |
| 0.67      | 10     | Light Gray |
| 1.00      | 11     | White      |

### Number of Gray Levels

| Bits   | Gray Levels |
|--------|-------------|
| 2 bits | 4           |
| 3 bits | 8           |
| 8 bits | 256         |

More bits produce smoother images.

### Grayscale Intensity Formula

For monochrome displays

$$\text{Intensity} = 0.5[\min(R, G, B) + \max(R, G, B)]$$

This converts an RGB color into a grayscale intensity.

**Question:** Calculate the grayscale intensity for **R = 0.9, G = 0.4, B = 0.6**.

**Solution:**

min = 0.4 , max = 0.9

$$\text{Intensity} = 0.5(0.4 + 0.9) = 0.5(1.3) = 0.65$$

### Advantages of Color Levels

- Realistic images
- Better visualization
- Easy object identification
- Millions of color combinations
- High-quality graphics

### Advantages of Grayscale

- Requires less memory
- Faster processing
- Suitable for printing
- Used in medical imaging
- Used in X-ray and CT images
- Useful for image processing

### Difference Between Color and Grayscale

| Color                  | Grayscale                   |
|------------------------|-----------------------------|
| Uses RGB colors        | Uses gray shades only       |
| Requires more memory   | Requires less memory        |
| Millions of colors     | Limited gray levels         |
| Realistic appearance   | Simple representation       |
| Used in color displays | Used in monochrome displays |

## 4. Color Attributes

**Definition:** A color attribute specifies the color used for drawing graphics. Computer graphics usually use the **RGB color model**.

RGB stands for:

- R = Red
- G = Green
- B = Blue

Every color is formed by mixing different intensities of these three primary colors.

### RGB Color Model

Each component ranges from:

$$0 \rightarrow 255$$

where:

- 0 = minimum intensity
- 255 = maximum intensity

Example:

Red

(255,0,0)

Green

(0,255,0)

Blue

(0,0,255)

## 5. Color Code Description Table (RGB)

| Color        | RGB Code        |
|--------------|-----------------|
| Black        | (0, 0, 0)       |
| White        | (255, 255, 255) |
| Red          | (255, 0, 0)     |
| Lime (Green) | (0, 255, 0)     |
| Blue         | (0, 0, 255)     |
| Yellow       | (255, 255, 0)   |
| Cyan         | (0, 255, 255)   |
| Magenta      | (255, 0, 255)   |
| Gray         | (128, 128, 128) |
| Orange       | (255, 165, 0)   |
| Pink         | (255, 192, 203) |
| Purple       | (128, 0, 128)   |
| Brown        | (165, 42, 42)   |

## 6. Character Attributes

Character attributes are properties that control the appearance of characters and text displayed on the screen. These attributes determine how text looks and improve readability.

### Main Character Attributes

#### 1. Font (Typeface)

A font is a collection of characters having the same design style.

##### Examples

- Times New Roman
- Arial

#### 2. Character Size

Controls the **height and width** of characters.

- Small size → more text fits on screen.
- Large size → better readability.

#### 3. Character Color

Determines the color of displayed characters.

Examples:

- Black
- Red

#### 4. Character Orientation

Controls the direction in which characters appear.

Possible orientations:

- Horizontal
- Vertical
- Rotated (45°, 90°, etc.)

## 5. Character Style

Characters can be displayed as:

- **Bold**
- *Italic*

These styles improve emphasis and appearance.

- Maps

## 7. Text Attributes

Text attributes control the appearance and positioning of an entire text string instead of individual characters. They determine how text is placed and aligned on the display.

### Main Text Attributes

#### 1. Text Font

Specifies the font used for the complete text.

Example: Hello World

#### 2. Text Height

Controls the overall height of the text.

Small: Hello

Large: HELLO

#### 3. Character Expansion Factor

Controls character width.

Normal: HELLO

Expanded: H E L L O

Compressed: HELLO

#### 4. Character Spacing

Controls the space between adjacent characters.

Example:

Normal: HELLO

Wide: H E L L O

#### 5. Text Path

Determines the direction in which characters are displayed.

Possible paths:

Left → Right

Right ← Left

Top ↓ Bottom

Bottom ↑ Top

#### 6. Text Alignment

Determines how text is aligned relative to its reference point.

Types:

- Left
- Center

- Right

## 7. Text Color

Specifies the color of the text.

Examples:

- Black
- Red

## 8. Pixel Phasing (Anti-aliasing)

Pixel Phasing, also known as Anti-Aliasing, is a technique used in computer graphics to reduce the jagged or stair-step appearance (aliasing) of lines and curves displayed on a screen.

### Why is Pixel Phasing Needed?

Raster displays produce jagged edges because pixels are square.

Example:

Without Anti-aliasing

#####

#####

#####

Edges appear rough.

### Basic Idea

Instead of drawing every pixel exactly at the grid center, the system shifts the displayed pixel by small fraction of a pixel. Typical shifts are:

- $1/4$  pixel
- $1/2$  pixel
- $3/4$  pixel

This makes the line appear closer to its true geometric position.

### How Pixel Phasing Works

1. Detect the actual line position.
2. Shift the displayed pixel slightly.
3. Adjust pixel intensity if needed.
4. The eye perceives a smoother edge.

### Advantages

- Reduces jagged edges.
- Produces smoother lines.
- Improves image quality.
- Makes diagonal lines appear more natural.

### Limitations

- Requires hardware support for sub-pixel positioning.
- Increases rendering complexity.
- Slightly increases computation.